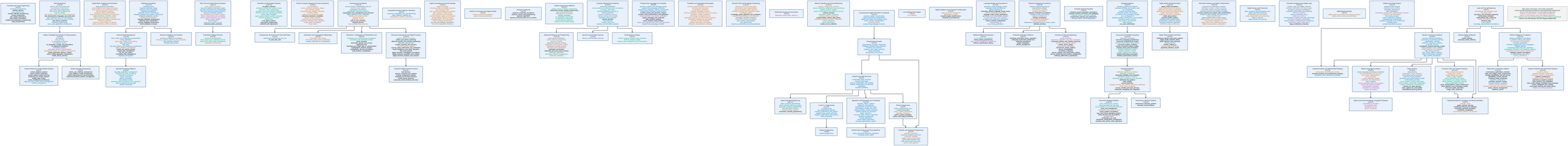

Functional and Logic Programming
9 ECTS
functional_programming
lambda_calculus
logic_programming
monadic_effectful_programming
monad_transformers
type_based_abstractions
type_system_foundations
certified_and_dependent_programming
scala_programming

Web Development
12 ECTS
internet_and_world_wide_web
web_access_infrastructure
web_client_side_programming
web_server_side_programming
web_api_design
web_solution_stacks_and_content_frameworks
web_development_languages_and_build_tools
web_application_and_xml_technologies
stateful_and_asynchronous_web_backends
web_service_protocols
javascript_programming
server_side_application_development

Digital Media, Graphics and Fabrication
12 ECTS
digital_graphics_foundations
raster_graphics_and_image_processing
vector_graphics_and_technical_drawing
cad_architectural_drawing
landscape_digital_representation
three_dimensional_modeling_and_fabrication
digital_fabrication_machine_control
digital_image_digitisation_formats
medical_3d_design_and_additive_manufacturing
immersive_platforms
immersive_technologies_for_fashion
digital_humanistic_content_production

Database Foundations
12 ECTS
database_basics
relational_database_design
relational_query_languages
database_transactions
database_system_architectures
database_storage_and_system_components
database_application_access
database_administration
relational_database_programming
relational_query_optimization
physical_database_performance
active_database_concepts

Real-Time and Cyber-Physical Systems
12 ECTS
cyber_physical_systems
real_time_systems_and_task_models
real_time_scheduling_and_schedulability
real_time_resource_access_protocols
real_time_operating_systems
cps_middleware_and_sensor_streams
sensor_measurement_and_modeling
sensor_signal_acquisition_and_quality
sensor_system_design
smart_sensors
motor_and_actuator_control
industrial_system_protocols

Simulation and Multi-Agent Systems
12 ECTS
simulation_methods
simulation_models_and_random_inputs
simulation_study_methodology
simulation_tools_and_parallel_execution
discrete_event_and_queueing_simulation
monte_carlo_simulation
agent_based_complex_system_simulation
machine_learning_in_simulation
digital_twins
multi_agent_systems
multi_agent_programming
collective_self_organization

Human-Computer Interaction and User Experience
12 ECTS
hci_foundations
human_centered_interaction_design
user_interface_design
prototyping_and_usability_evaluation
usability_testing
usability_and_accessible_design
advanced_interaction_modalities
service_design_methods
immersive_environment_design
immersive_hardware_and_interaction
mixed_reality_foundations
virtual_reality

Cybersecurity Foundations
12 ECTS
security_threats_vulnerabilities_and_risk
common_cyber_attacks
cryptography_foundations
cryptographic_security_properties
cryptographic_key_management_and_protocols
digital_signatures_and_certificates
identity_and_access_control
network_security
application_security
data_at_rest_protection
physical_security
basic_cybersecurity_practices

Geospatial and Data Collection Laboratory
6 ECTS
numerical_cartography_and_georeferencing
data_collection_analysis_lab
spreadsheet_workflow_automation

Digital Humanities and Cultural Heritage
12 ECTS
digital_humanities
humanities_data_and_digital_resources
fair_and_open_data
open_science_principles_and_methods
open_science_infrastructures
open_scholarly_communication
cultural_heritage_digital_communication
archaeological_digital_surveying
photogrammetric_3d_reconstruction
virtual_heritage_reconstruction
gis_and_web_heritage_applications
document_digitization_workflows

Fashion E-Commerce and Digital Visibility
3 ECTS
fashion_ecommerce_platforms
search_engine_optimization

Quantum Computing
6 ECTS
quantum_computing_foundations
quantum_circuits_and_programming
quantum_algorithms
quantum_protocols_and_error_correction

Parallel Programming Platforms
9 ECTS
shared_memory_parallel_programming
distributed_memory_parallel_programming
gpu_simd_programming
heterogeneous_computing
parallel_computing
ml_workloads_and_kernels
ml_algorithmic_optimization
ml_acceleration_platforms
efficient_neural_model_design

Computer Networking Foundations
12 ECTS
networking_foundations
communication_and_physical_layer_foundations
link_layer_networking
internet_architecture_and_tcp_ip
internet_protocols
transport_reliability_protocols
internet_routing_and_multicast
application_layer_protocols
network_services
web_protocols
wireless_networking
wireless_communication_systems

Programming Languages and Compilers
12 ECTS
grammar_notation_and_language_syntax
programming_language_semantics
programming_language_runtime_environments
programming_language_implementation
compiler_lexical_and_semantic_support
compiler_code_transformation_and_generation
formal_grammars_and_parser_generation
regular_languages_and_automata
static_program_analysis
virtual_machine_implementation
model_driven_development_and_code_generation
programming_language_evolution

Translation and Interpreting Technologies
12 ECTS
computer_aided_translation_foundations
cat_tools_and_resources
ai_translation_assistance
machine_translation_systems
machine_translation_preparation_and_post_editing
translation_quality_evaluation
terminology_management_tools
specialized_corpora_and_terminology_extraction
localization_engineering_and_quality
localized_content_translation
software_localization_foundations
interpreter_documentation_strategies

Semantic Web and Knowledge Engineering
12 ECTS
semantic_web_and_linked_data
rdf_semantic_web_representation
sparql_semantic_knowledge_graph_querying
ontology_modeling_and_description_logics
knowledge_graphs
knowledge_graph_alignment_and_completion
semantic_representation_and_inference
knowledge_extraction_methods
information_retrieval_foundations
web_content_representation
json_data_format
electronic_documents_and_dematerialization

Distributed Immersive Environments
3 ECTS
distributed_virtual_reality_platforms

Network Operations and Virtual Networking
12 ECTS
network_configuration_protocols
network_interface_configuration_and_monitoring
network_protocol_analysis
network_performance_and_quality_of_service
network_proxying
centralized_system_monitoring
software_defined_networking
programmable_network_data_planes
telecommunications_network_virtualization
virtual_networking
virtual_networking_2
mobile_ad_hoc_networks

Computing and Digital Information Foundations
6 ECTS
computer_system_fundamentals
digital_information_representation
digital_information_and_software_levels
binary_number_representation
computing_history
programming_language_evolution

Live Subtitling Technologies
3 ECTS
live_subtitling

Digital Intelligence and Industrial Transformation
6 ECTS
digital_data_and_intelligence
digitalization_impacts
industrial_digital_transformation
industrial_iot_architectures

Language Models and Generative AI
12 ECTS
generative_large_language_models
language_model_architectures
prompt_engineering
parameter_efficient_language_model_tuning
retrieval_augmented_generation
language_model_action_applications
generative_ai_for_knowledge_extraction
ai_code_generation_and_synthesis
ai_safety_and_preference_optimization
human_machine_collaboration
neurosymbolic_ai
text_mining_applications

Software Engineering Foundations
12 ECTS
software_engineering_lifecycle
requirements_analysis
uml_software_modeling
software_architecture
software_architecture_foundations
software_quality_and_testing
software_maintenance
software_prototyping
agile_software_development
project_management
collaborative_development_and_version_control
software_licensing

Computer Science Education
6 ECTS
computer_science_education_foundations
computer_science_curriculum_and_pedagogy
computer_science_teaching_methods
programming_learning_and_assessment
scholarly_search_and_bibliographies

Operating Systems
12 ECTS
operating_systems_foundations
processes_threads_and_system_calls
cpu_scheduling
process_synchronization
operating_system_file_systems
os_memory_and_secondary_storage
os_io_management
operating_system_interprocess_communication
operating_system_security
local_system_administration
shell_scripting
unix_user_authorization

Digital Health and Bioinformatics
12 ECTS
digital_health_foundations
health_data_privacy_and_security
healthcare_information_applications
healthcare_document_and_data_lifecycle
health_information_systems
healthcare_information_workflows
health_data_science
personal_health_systems
biological_data_representation
bioinformatics_sequence_analysis
biological_sequence_algorithms
biomedical_signal_processing

Information Systems and Digital Transformation
12 ECTS
organizational_information_systems
information_system_initiative_planning
information_system_process_modeling
business_process_and_workflow_modeling
business_process_and_master_data_transformation
enterprise_data_strategy_and_governance
digital_transformation
digital_transformation_project_design
digital_identity_trust_and_public_services
documentary_database_use
digital_rights_and_intellectual_property
digital_social_decision_implications

Digital Literacy and Productivity
6 ECTS
basic_computer_and_productivity_tools
digital_document_authoring
spreadsheet_data_processing
online_collaboration_and_social_web
basic_it_security
digital_privacy_and_security

Computer Architecture and Digital Logic
12 ECTS
boolean_logic_and_digital_networks
hardware_description_and_logic_design
processor_instruction_architecture
processor_performance_and_pipelining
memory_hierarchy
memory_interrupts_and_io
system_bus_dma_io
microprocessor_systems_and_chip_technology
processor_protection_virtualization
shared_memory_multiprocessor_architecture
architecture_power_performance
fpga_accelerated_computing

Digital Media Industries
3 ECTS
digital_media_economics_and_industry_structure
digital_media_industry_transformation

Statistics and Data Analysis
12 ECTS
statistical_data_analysis
statistical_study_methods
probability_distributions
data_visualization
multivariate_analysis_and_dimensionality_reduction
similarity_and_dissimilarity
time_series_analysis
python_descriptive_analytics
r_programming_fundamentals
r_graphics_packages_and_randomness
analytics_types
variable_importance_analysis

Logic and Formal Reasoning
9 ECTS
propositional_and_syllogistic_logic
first_order_logic
set_theory_and_structural_reasoning
formal_logic_motivation
mathematical_logic_reasoning
modal_and_action_logic
temporal_logic
event_calculus
knowledge_representation_foundations

Topic colour: first scope, in this order, among the
current courses teaching it (by teachers' and programmes' department)
service: some DISI teacher, no DISI degree programme
external: no DISI teacher, no DISI degree programme
borrow: no DISI teacher, some DISI degree programme
weak_internal: some (not all) DISI teachers and DISI degree programmes
internal: only DISI teachers and DISI degree programmes

Mobile, Embedded and Internet of Things Systems
12 ECTS
iot_systems
web_of_things
iot_messaging_protocols
iot_network_routing
iot_application_models_and_descriptions
iot_deployment_platforms
embedded_systems
embedded_device_communications
embedded_firmware_and_platform_development
mobile_application_delivery_models
native_mobile_application_development
mobile_device_services

Advanced Data Management
12 ECTS
data_models_and_multimedia_representation
nosql_databases
nosql_data_models_and_sharding
data_warehousing
data_lakes
big_data_platform_and_lakehouse_architectures
graph_database_management
ranked_and_skyline_queries
spatial_data_management
sql_xml_and_xquery
multimedia_data_management
database_transactions

Business Intelligence and Analytics
6 ECTS
business_intelligence_foundations
business_analytics_reporting_and_prescription
data_analytics_practice
financial_data_access

Embedded Intelligent Devices
6 ECTS
electronics_fundamentals
smart_spaces_and_intelligent_devices
sensor_time_series_data_management
embedded_cybersecurity

Cybersecurity Governance and Dark-Web Risks
3 ECTS
cybersecurity_governance_and_risk
tor_and_dark_web

Information and Cryptographic Mathematics
6 ECTS
information_and_entropy
information_theory_data_compression
modular_arithmetic_for_cryptography

Blockchain, Cryptography and Distributed Trust
9 ECTS
blockchain_and_cryptocurrency_foundations
blockchain_and_distributed_ledgers
blockchain_security_and_privacy
smart_contracts
distributed_trust
federated_and_single_sign_on_authentication
cryptographic_protocol_analysis
cryptographic_security_foundations
post_quantum_cryptography

Advanced Cybersecurity and Digital Forensics
12 ECTS
network_security_monitoring
system_and_device_hardening
offensive_security_and_reverse_engineering
defensive_cybersecurity
incident_response_and_recovery
risk_and_security_auditing
security_policy_frameworks_and_certification
threat_intelligence_and_cyber_deception
zero_trust_security
digital_forensics_investigation_foundations
digital_evidence_acquisition_and_custody
digital_forensics_analysis_and_recovery

Networked Systems and Programming
12 ECTS
client_server_architecture
socket_network_programming
remote_invocation
distributed_resource_naming
distributed_communication_models
distributed_message_passing
distributed_data_serialization
distributed_coordination
distributed_time_and_event_ordering
distributed_algorithms
distributed_resource_management
distributed_scalability

Internet Communication Services
3 ECTS
internet_communication_services

Social Network Analysis
12 ECTS
network_analysis_concepts_and_measures
network_research_design
network_data_management_and_visualization

Programming Concepts
6 ECTS
programming_concepts_introduction
imperative_programming_constructs
functions_and_recursion
program_input_output_and_files
string_processing
development_environment

Software Platform Development
6 ECTS
ai_in_software_development
csharp_dotnet_programming
kotlin_and_multiplatform_programming
behavior_specification_testing

Enterprise Application Platforms
6 ECTS
enterprise_application_server_operations
enterprise_component_frameworks
enterprise_persistence
system_integration
identity_management

DevOps and Service Engineering
12 ECTS
devops_practices
continuous_integration_and_build_automation
continuous_delivery_and_release_versioning
test_first_and_property_testing
domain_driven_design
microservices_architecture
microservice_design_patterns
service_choreography
serverless_computing
service_capacity_and_performance_testing
service_monitoring_and_incident_management
software_performance_engineering

Concurrent and Parallel Computing
12 ECTS
concurrent_computing
concurrent_language_programming
concurrent_programming_models
asynchronous_programming_models
concurrent_system_modeling
formal_concurrent_system_models
process_calculi_and_bisimulation
parallel_architecture_models
parallel_algorithm_models
parallel_programming_patterns
parallel_performance_evaluation
software_transactional_memory

Digital Health Systems and Ethics
9 ECTS
healthcare_delivery_and_payer_systems
digital_health_research_ethics
health_behavior_and_outcomes
medical_devices
medical_imaging_ai
medical_image_processing
systematic_literature_review

Machine Learning Foundations
12 ECTS
machine_learning_foundations
supervised_unsupervised_learning
classification_methods
predictive_regression_analytics
ensemble_learning
probabilistic_machine_learning_models
model_evaluation_and_validation
model_selection_and_validation
data_mining_overview
data_mining_pattern_discovery
data_mining_project_workflow
data_science_foundations

Software Model Verification
3 ECTS
model_checking
advanced_model_checking

Artificial Intelligence Foundations
12 ECTS
artificial_intelligence_foundations
ai_problem_solving_and_search
ai_planning_and_scheduling
constraint_satisfaction
constraint_programming_and_propagation
intelligent_agents
knowledge_based_and_rule_reasoning
probabilistic_reasoning_with_bayesian_networks
automated_decision_system_architectures
intelligent_systems_foundations
ai_epistemology_and_ethics
robotics_foundations

Context-Aware and Location-Based Systems
9 ECTS
context_adaptive_systems
human_activity_recognition
location_based_context_services
positioning_for_mobile_systems
spatial_data_privacy
mobile_middleware_architectures
edge_fog_and_mobile_service_management
mobility_management

Mobile Application Engineering
6 ECTS
mobile_user_interface_development
cross_platform_mobile_development
mobile_networking_and_synchronization
telecommunications_session_management

Big Data Processing Platforms
9 ECTS
big_data_application_management
clustered_big_data_processing
mapreduce_computing
spark_computing
hadoop_ecosystem
data_pipeline_design
distributed_data_consistency_and_replication
cloud_computing_for_big_data
parallel_computing

Advanced Digital Forensics Practice
9 ECTS
anti_forensics
forensic_command_line_systems
forensic_professional_practice
virtualized_environment_forensics
mobile_device_forensics
networked_communications_forensics

Programming Data Structures
6 ECTS
fundamental_data_structures
abstract_data_types
memory_organization_and_pointers
program_organization_and_libraries
debugging
exception_handling

Distributed Systems
12 ECTS
distributed_systems_foundations
distributed_middleware
distributed_reliability_and_availability
distributed_quality_of_service
distributed_data_consistency_and_replication
distributed_file_systems
code_mobility
service_discovery
message_oriented_and_publish_subscribe_middleware
event_driven_computing
service_oriented_and_web_services
enterprise_messaging_and_integration

Industrial Analytics and Mathematical Modeling
6 ECTS
mathematical_modeling_for_industry
industrial_anomaly_and_maintenance_analytics
operational_and_maintenance_analytics

Natural Language Processing
12 ECTS
natural_language_processing_foundations
text_preprocessing_and_nlp_tools
text_vector_representations
corpus_analysis
n_gram_language_models
supervised_text_classification
topic_modeling
natural_language_syntax_and_parsing
lexical_and_logical_semantics
computational_linguistics
computational_morphology
speech_processing

Deep Learning
12 ECTS
deep_learning
feedforward_neural_networks
convolutional_neural_networks
recurrent_neural_networks
generative_and_autoencoding_models
reinforcement_learning
neural_network_training_optimization
neural_network_robustness_interpretation
model_compression_and_efficiency
calculus_for_deep_learning
linear_algebra_for_computation
computational_learning_theory

Computer Vision and Image Processing
12 ECTS
digital_image_fundamentals
image_color_models
image_filtering_and_edges
binary_image_analysis
camera_geometry_and_stereo
local_features_and_shape_matching
object_detection_recognition_tracking
vision_classification_training
visual_representations_metric_transformers
vision_attention_and_multimodal_models
monocular_depth_estimation
image_video_captioning

Optimization and Decision Making
12 ECTS
combinatorial_optimization_methods
linear_and_integer_linear_programming
boolean_satisfiability_and_sat_solving
satisfiability_modulo_theories
constraint_based_scheduling
algorithm_configuration
stochastic_decision_models
business_decision_modeling
decision_quality_monitoring
industrial_production_and_resource_planning
power_network_management
epidemic_control

Cognitive Robotics and Autonomous Systems
12 ECTS
cognitive_science
semiotic_semantic_frameworks
multimodal_cognitive_interpretation
cognitive_architectures
intelligent_systems_engineering
autonomous_adaptive_systems
robot_behavior_and_adaptation
robot_control_design_navigation_and_evaluation
swarm_intelligence_and_robotics
multi_agent_learning_and_game_theory
robotics_autonomous_systems

Object-Oriented Programming
6 ECTS
object_oriented_programming
object_oriented_design_principles
java_data_structure_programming
java_application_features
software_design_patterns
component_oriented_programming

C and C++ Programming
6 ECTS
c_programming
cpp_programming_features
memory_organization_and_pointers
program_input_output_and_files
program_organization_and_libraries
string_processing

Algorithms, Data Structures and Complexity
12 ECTS
algorithm_design_analysis
algorithm_cost_analysis
computational_complexity_theory
computability_models_and_limits
sorting_selection_priority_structures
hashing_disjoint_set_structures
graph_algorithms
recursive_divide_conquer_algorithms
dynamic_programming
greedy_algorithms
exact_search_techniques
heuristic_approximation_search

Python Programming
6 ECTS
python_programming_core
scripting_languages
python_regular_expressions
notebook_computing
python_project_packaging
python_web_page_processing

Cloud and Virtualized Systems
12 ECTS
cloud_infrastructure_models
cloud_computing_for_big_data
cloud_migration_and_resource_federation
cloud_cost_management
virtualization_and_cloud_services
virtual_systems_foundations
user_level_virtual_operating_systems
virtual_devices_and_file_systems
containerization
configuration_as_code
centralized_configuration_management
container_and_service_mesh_operations

Autonomous Vehicular Systems
3 ECTS
autonomous_vehicular_systems
vehicular_communications

Speech and Advanced Natural Language Processing
6 ECTS
computational_phonetics
nlp_semantic_tasks
probability_for_nlp
speech_recognition

Multimedia Retrieval, Annotation and Recommendation
9 ECTS
computer_vision
graph_machine_learning
multimedia_retrieval_and_feedback
multimedia_semantic_annotation
multimedia_visualization
recommender_systems_and_fairness

System Programming
3 ECTS
system_programming

Bioinformatics Search and String Algorithms
3 ECTS
string_processing_structures_algorithms
biological_search_tasks

Scientific and Numerical Programming
9 ECTS
scientific_computing
numerical_analysis_and_errors
numerical_modeling
matlab_octave_programming
python_data_science_tools
open_source_and_linux_tools
python_turtle_graphics